



DRONE

CLOUD SEEDING

Introduction

Drone cloud seeding is a modern technology that involves the use of unmanned aerial vehicles to disperse seeding agents into clouds in order to induce rainfall. This innovative approach offers a more cost-effective and efficient method of weather modification compared to traditional ground-based methods.



PROCESS OF CLOUD SEEDING



1. Identifying Target Clouds

~ Clouds suitable for seeding are identified. These are typically clouds with enough moisture but insufficient natural process to trigger precipitation.

2. Seeding Agents

~ Substances like silver iodide, salt, dry ice are used. These act as "nuclei" around which water vapour can condense.

3. Dispersal Methods

~ Drones are equipped with flares containing seeding materials that are burned and released directly into clouds.

4. Seeding Clouds

~ The seeding agents are dispersed into the cloud. In the case of Silver Iodide, the particles encourage the condensation of water vapour around them, forming ice crystals or water droplets.

5. Formation of Ice Crystals

~ In the cloud , the seeding agents help water droplets or ice crystals form. These droplets combine and grow larger. when the droplets or ice crystals become heavy enough, they fall as rain or snow, depending on the temperature.

6. Outcome

~ This method can either increase rainfall in drought -prone areas or help control the timings of precipitation in certain regions.

DRONES USED

- DJI Phantom 4
- Bell autonomous pods transport (APT)
- DJI Matrice 600
- FreeFly systems Astro



Benefits of Using Drones for Cloud Seeding



Reduced risk

The use of drones for cloud seeding reduces the need for human pilots to fly through potentially hazardous weather conditions. This decreases the risk of accidents and ensures the safety of personnel.

Efficiency and precision

Using drones for cloud seeding offers increased precision and accuracy in targeting specific areas for rainfall. This can result in more efficient water distribution and better crop yields.

Accessibility and safety

Drones are able to access hard-to-reach or dangerous areas that traditional methods may not be able to reach. This allows for cloud seeding to be conducted in a wider range of locations.

Conclusion: The Role of Drones in Enhancing Cloud Seeding Efforts

Accessibility of drones



Drones have the potential to greatly enhance cloud seeding efforts by providing more precise and targeted delivery of seeding materials {Advantages of drones}. With their ability to navigate through difficult terrain and reach remote locations, drones can help reach clouds that are unreachable by traditional methods .

Enhanced success rate



Furthermore, drones offer a cost-effective solution for cloud seeding operations, reducing the need for expensive aircraft and manpower {Cost-effective solution}. By improving the accuracy and efficiency of cloud seeding, drones can help increase the success rate of precipitation enhancement programs .



THANK YOU